Innovation Diffusion Analysis: TIME 2025 Robotics

Course Assignment: Bass Diffusion Model Analysis

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1. Selected Innovation

Innovation: Figure 03 Household Humanoid Robot (Robotics Category)

TIME 2025 Link: https://time.com/collections/best-inventions-2025/7318507/figure-03/

2. Historical Look-Alike Innovation & Justification

Look-Alike Innovation: Consumer Robotic Vacuum Cleaners (iRobot Roomba)

The adoption trajectory of modern household humanoids (such as Figure 03) closely mirrors early consumer robotic vacuums. Functionally, both categories shift households from manual human labor to autonomous physical task execution. Technologically, both depend on onboard spatial sensors (LiDAR, computer vision) and hardware unit-cost reductions before achieving broad market adoption. Economically, both faced early high hardware costs and consumer skepticism regarding daily utility.

3. Data Source

Historical Reference: Statista Research / iRobot Historical Unit Sales Data (2005â€“2020).

4. Estimated Bass Model Parameters

- **Coefficient of Innovation (p):** 0.006426
- **Coefficient of Imitation (q):** 0.323217
- **Historical Market Capacity (M_hist):** 147.08 Million units

5. Scope Selection & Justification

Selected Scope: Global Market

Humanoid robotics platforms require substantial R&D and capital expenditures, making international markets essential from launch.

6. Fermi Estimation for Target Market Potential (M)

To derive the global market capacity (M) for household humanoids:

1. Total middle/high-income global households = 500,000,000
2. Addressable target adoption tier (20%) = 100,000,000 households
3. Estimated 15-year penetration limit (50%) = 100,000,000 × 0.50 = **50,000,000 units** (M).

7. Diffusion Predictions

The predicted 15-year adoption path is exported to `data/predicted_diffusion.csv` and plotted in `img/image1.png`.